INSTALLATION CHECKLIST
EMS BASIC

PROJECT NAME: .....

PROJECT ADDRESS: .....

CONTACT PERSON: .....

Point of contact for sending the checklist: projekty@energostation.com

In case of questions please contact: Ing. Barbora Říčková

Tel. : +420 776 343 150

barbora.rickova@energostation.com

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Introductory Information

Each item in the Checklist is explained in a separate document sent together with the checklist.

The checklist uses several methods for filling in answers to each item.

a) $\square \rightarrow \boxtimes$ - indicates the need to check off and complete the item.

b) **YES NO** $\rightarrow$ **YES ~~NO~~** - indicates the need to select one option; please cross out the invalid choice.

c) **A) B) ... X)** $\rightarrow \ldots {}^{\mathbf{B}} \ldots$ - indicates the option to choose from the listed options and enter the letter of the selected choice on the dotted line at the end of the section.

d) ..... - indicates the need to fill in a value for the given item.

If a particular item does not apply to your technology — for example, if inverters use MODBUS TCP connection, do not fill in the section dealing with inverter connection via MODBUS RTU or Logger.

Installation Checklist

1.1 Electrical Installation Work

1. Installation and power connection of the EMS/CS panel. YES NO
2. Location of the EMS/CS panel.[1] INDOORS OUTDOORS
3. Installation and power connection of the auxiliary M&C panel(s). YES NO
4. Number of actually installed M&C panels. .....
5. Installation and power connection of the Remote Metering panel. YES NO
6. Labelling of panels according to SPD[2]

Panel name	Panel label per SPD
EMS / CS Panel	

1.2 Electricity Meters

1.2.1 SML133 Electricity Meter in EMS/CS Panel

a) Electricity meter installed for measuring:

A) At the delivery point with D.S.
B) Production module (PV).
C) Production module (BESS).
D) Other: .....

Selected measurement mode .....

b) Installation and connection of CT (current transformer). ☐

c) Installation and connection of VT (voltage transformer).[3] ☐

d) Verification of correct current flow direction on CT. ☐

[1] Specify the location — whether the panel is installed outdoors or placed in a substation.
[2] If multiple panels are installed, use the notes field.
[3] Fill in only if the meter will measure at medium voltage (MV) level.

e) Phase sequence check. ☐

f) Voltage-to-current assignment check. ☐
($I_1 \rightarrow U_1; I_2 \rightarrow U_2; I_3 \rightarrow U_3$)

g) Meter configuration:

- CT ratio setting. YES NO
- CT value. .....
- Measured voltage level set. YES NO
- Voltage level value. .....
- VT ratio set.[4] YES NO
- VT value. .....

1.2.2 Sub-meters

a) Number of installed electricity meters, e.g. for measuring PV, sub-meters:

.....

b) Information on additional installed sub-meters:

Label per SPD	Meter type	Modbus ID IP address :: ID	Ratio CT	Terminal block[a]
		.... : .... : .... : .... ::		X1 / X2
		.... : .... : .... : .... ::		X1 / X2
		.... : .... : .... : .... ::		X1 / X2
		.... : .... : .... : .... ::		X1 / X2
		.... : .... : .... : .... ::		X1 / X2

[a] Cross out whichever does not apply.

c) Notes on electricity meters:

.....

.....

.....

[4] Fill in only if the meter will measure at medium voltage (MV) level.

.....

.....

.....

1.3 Parametrisation and Communication

a) Connection of Ethernet cable (data network) to the EMS/CS panel. ☐

b) Connection of Ethernet cable (data network) to all M&C panels. ☐

c) Data network configuration for communication of control units and the following exceptions in FireWall rules: ☐

- vpn.icool.cloud tcp/1194
- cz.pool.ntp.org
- 8.8.8.8 (DNS)
- www.pvforecast.cz
- www.ote-cr.cz

d) IP addresses and IC numbers of control units. [5]

IC number of control unit	IP Address	Addressing[a]
	.... : .... : .... :	STATIC / DHCP
	.... : .... : .... :	STATIC / DHCP
	.... : .... : .... :	STATIC / DHCP

[a] Cross out whichever does not apply.

e) Parameters of the local network to which the technology is connected:

- Network IP address: ..... : ..... : ..... : .....
- Subnet mask: ..... : ..... : ..... : .....
- Gateway: ..... : ..... : ..... : .....
- DNS: ..... : ..... : ..... : .....
- All components using TCP connection (inverters, batteries, etc.) are mutually reachable via PING command. ☐

f) Method of connecting the installation site / EMS/CS unit to the Internet:

A) Connection to local data network.

B) Connection via mobile operator.

Selected connection mode: .....

Configuration may only begin once data connectivity is functional! Attach a photo of the PLC display showing confirmation of correct PLC-to-cloud communication; status [100] is required. See chapter 3.1.

[5] When taking photo documentation of IC numbers, remember to label the photo with the corresponding panel.

1.4 Technology

1.4.1 Inverters Connected via Logger

a) Connected to the data network with visibility to EMS/CS. ☐

b) Logger control by superior system enabled. ☐

c) All controlled inverters connected. ☐

Logger type	Total inverter power [kW]	PV power [kWp]	IP Address	Modbus ID	Firmware version
			.... : .... : .... : .... port:		
			.... : .... : .... : .... port:		
			.... : .... : .... : .... port:		

1.4.2 Inverters Connected via Modbus RTU

a) Installation of 120Ω termination resistors at each end of every communication line. ☐

b) Verification of RS-485 line polarity (A, B, GND). ☐

c) Inverter control via Modbus enabled. ☐

d) All controlled inverters connected. ☐

Label per SPD	Manufacturer and type	Power [kW]	PV power [kWp]	Modbus ID	Line speed [bps]

1.4.3 Inverters Connected via Modbus TCP

a) Connected to the data network with visibility to EMS/CS. ☐

b) Inverter control via Modbus enabled. ☐

c) All controlled inverters connected. ☐

Label per SPD	Manufacturer and type	Power [kW]	PV power [kWp]	IP Address	Modbus ID
				.... : .... : .... : .... port:	
				.... : .... : .... : .... port:	
				.... : .... : .... : .... port:	
				.... : .... : .... : .... port:	
				.... : .... : .... : .... port:	
				.... : .... : .... : .... port:	
				.... : .... : .... : .... port:	
				.... : .... : .... : .... port:	

1.4.4 Battery Storage Systems

a) All battery storage system components connected to EMS/CS. [6] ☐

b) Information on battery storage and component addressing:

- Battery storage type / model: .....
- Storage capacity: ..... [kWh]
- Max. SoC for charging: ..... %
- Min. SoC for discharging: ..... %

c) Additional information on battery storage:

[6] Battery inverters, BMS, battery units, etc.

Component INV / BMS / MBMS	BS capacity [kWh]	Max. inverter power [kW]	Modbus ID IP Address :: ID	Storage ID [a]
			.... : .... : .... : .... ::	
			.... : .... : .... : .... ::	
			.... : .... : .... : .... ::	
			.... : .... : .... : .... ::	
			.... : .... : .... : .... ::	
			.... : .... : .... : .... ::	

[a] In case of MBMS.

d) Notes on batteries:

.....

.....

.....

.....

.....

.....

1.5 Weather Station

a) METEO station installed with regard to measuring the irradiation of the power plant[7]. ☐

b) IP address of the METEO station is on the same network as the EMS/CS. [8] ☐

c) IP address of the METEO station: .... : .... : .... : ....

1.6 Compensation

a) Compensation installed? YES NO

b) Will it affect PV control and commands from CS? YES NO

c) Does the compensation unit support blocking via contact? YES NO

[7] Values from the METEO station measurements are transmitted to the dispatch centre. It is therefore necessary that they correspond to actual irradiance, wind speed, etc.

[8] Mutually reachable via PING command.

System Settings

For the purpose of configuring the EMS/CS, several basic parameters of the energy management service itself, the serviced devices (inverter, battery storage) and controlled consumers (proportionally controlled coil of a thermal storage tank, e-mobility) need to be filled in. Based on the entered values, the EMS control system will be configured and handed over.

2.1 Consumption Point Parameters

a) Nominal value of the **main circuit breaker**: $I_N =$ ..... [A]

- For MV, convert to equivalent: $I_{N-MV} =$ ..... [A]

b) Value of **reserved input** per connection agreement (SoP): ..... [kW]

c) Value of **reserved output** of the generation unit per SoP: ..... [kW]

d) Total **installed capacity** per SoP: ..... [kWp]

e) Permission to export power to the grid during trial operation: YES NO

- Permitted export value: ..... [kW]

2.2 SPOT / SELL Parameters

Electricity buy-out is realised according to prices:

A) SPOT - prices according to the OTE exchange.

B) FIX - fixed price agreed with the trader.

Selected electricity sell mode .....

a) **SPOT**

- Transmission system usage fee: ..... [EUR / MWh]
- VAT rate: ..... [%]
- Minimum profit: ..... [%]
- Disable grid export at negative SPOT prices: YES NO

b) **FIX**

- Fixed selling price: ..... [EUR / MWh]
- Transmission system usage fee: ..... [EUR / MWh]
- VAT rate: ..... [%]

2.3 SPOT / PURCHASE Parameters

Electricity purchase is realised according to prices:

A) SPOT - prices according to the OTE exchange.

B) FIX - fixed price agreed with the trader.

C) SPOT+FIX - OTE exchange prices up to the agreed price cap.

Selected electricity purchase mode .....

a) **SPOT**

- Costs at high tariff: ..... [EUR / MWh]
- Costs at low tariff: ..... [EUR / MWh]
- Daily costs: ..... [EUR / MWh]
- VAT rate: ..... [%]

b) **FIX**

- Fixed purchase price: ..... [EUR / MWh]
- Costs at high tariff: ..... [EUR / MWh]
- Costs at low tariff: ..... [EUR / MWh]
- Daily costs: ..... [EUR / MWh]
- VAT rate: ..... [%]

c) **SPOT+FIX**

- Maximum purchase price: ..... [EUR / MWh]
- Costs at high tariff: ..... [EUR / MWh]
- Costs at low tariff: ..... [EUR / MWh]
- Daily costs: ..... [EUR / MWh]
- VAT rate: ..... [%]

!!! IMPORTANT: if the battery will be charged from SPOT and the consumption point is billed according to the 1/4h reserved capacity (RC), it is necessary to address the monitoring and control of 1/4h power directly from the billing meter. If 1/4h reserved capacity control from the billing meter is not used, the RC limit may be exceeded and the consumption point may be billed for exceeding the RC limit.

2.4 Parameters of Reserved Capacity 1/4h Maximum

a) Reserved capacity 1/4h maximum: ..... [kW]

b) Galvanic opto-isolator for pulse collection from the billing meter installed and connected: YES NO

c) Galvanic opto-isolator connected to the 1/4h maximum panel: YES NO

d) 1/4h maximum panel connected to EMS/CS panel: YES NO

- CIB address of AK.PULS element (C-AM-0006I): .....

e) Pulse description:

- imp1 (AK.PULS/AI1): .....
- imp2 (AK.PULS/AI2): .....
- imp3 (AK.PULS/AI3): .....

f) Control stages:

- Stage 1 – battery compensation: YES NO
- Stage 2 – device 1 shutdown: max. input ..... [kW]
- Stage 3 – device 1 shutdown: max. input ..... [kW]
- Stage 4 – device 1 shutdown: max. input ..... [kW]
- Stage 5 – device 1 shutdown: max. input ..... [kW]
- Additional stages must be addressed individually.

2.5 Battery Storage Settings

The battery storage system can be operated in one of the following modes (combining modes always needs to be agreed individually, after verifying that the given technology supports the combination).

A) **Self-consumption**: charging from PV surplus and consumption compensation.

B) **Peak-shaving**: trimming/levelling power peaks (standard consumption is not compensated).

C) **SPOT**: charging and compensation based on SPOT prices (does not handle power peak balancing).

Selected battery storage control mode .....

2.6 EMS Source Settings for SPOT

Control of PV sources, battery storage or other EMS sources. If a source, such as a battery, will not be operated in SPOT mode, do not fill in this section.

a) Enable EMS control according to SPOT: YES NO

b) Battery storage:

- Enable battery discharge to the grid according to SPOT: YES NO
- Battery amortisation when using SPOT: ..... [EUR / MWh]
- Battery storage losses: ..... [%]

c) Are additional sources installed in EMS/CS?[9] YES NO

Notes:

[9] For example, cogeneration units or diesel generators. Describe these sources in the notes below.

.....

.....

.....

.....

.....

.....

Supporting Documents for Checklist

3.1 Mandatory Attachments

a) Photo documentation of control unit displays (IC numbers). ☐

b) Photo documentation of nameplates of controlled technologies as listed above. ☐

c) Technical data sheets for the technologies used. ☐

d) Modbus register table according to the currently installed firmware and interface type (TCP/RTU).[10] ☐

e) Panel documentation with the implemented signals and commands marked, including any recorded changes. ☐

f) As-built project documentation:

- Single-line diagram showing the complete affected technology and affected panels (incl. elements per the telemetry table). ☐
- Site layout diagram showing the location of affected technologies and panels. ☐

g) Connection agreement in its current version, including appendices and amendments. ☐

h) Screenshot (photo of screen) from the distribution portal showing the contracted values of reserved input, output, and capacity. ☐

By signing, I confirm the accuracy of the information provided and the verifications carried out. I acknowledge that if any discrepancy is found during configuration and functional testing, the resolution of such discrepancy will be handled by EnergoStation s.r.o. as additional work charged at the hourly rate from the current price list.

In ..... on: .....

Checklist completed by (legibly): Company:

....

.....

SIGNATURE

[10] Only if the installed technologies include a device that is not supported by the EMS/CS system and a new driver needs to be programmed.